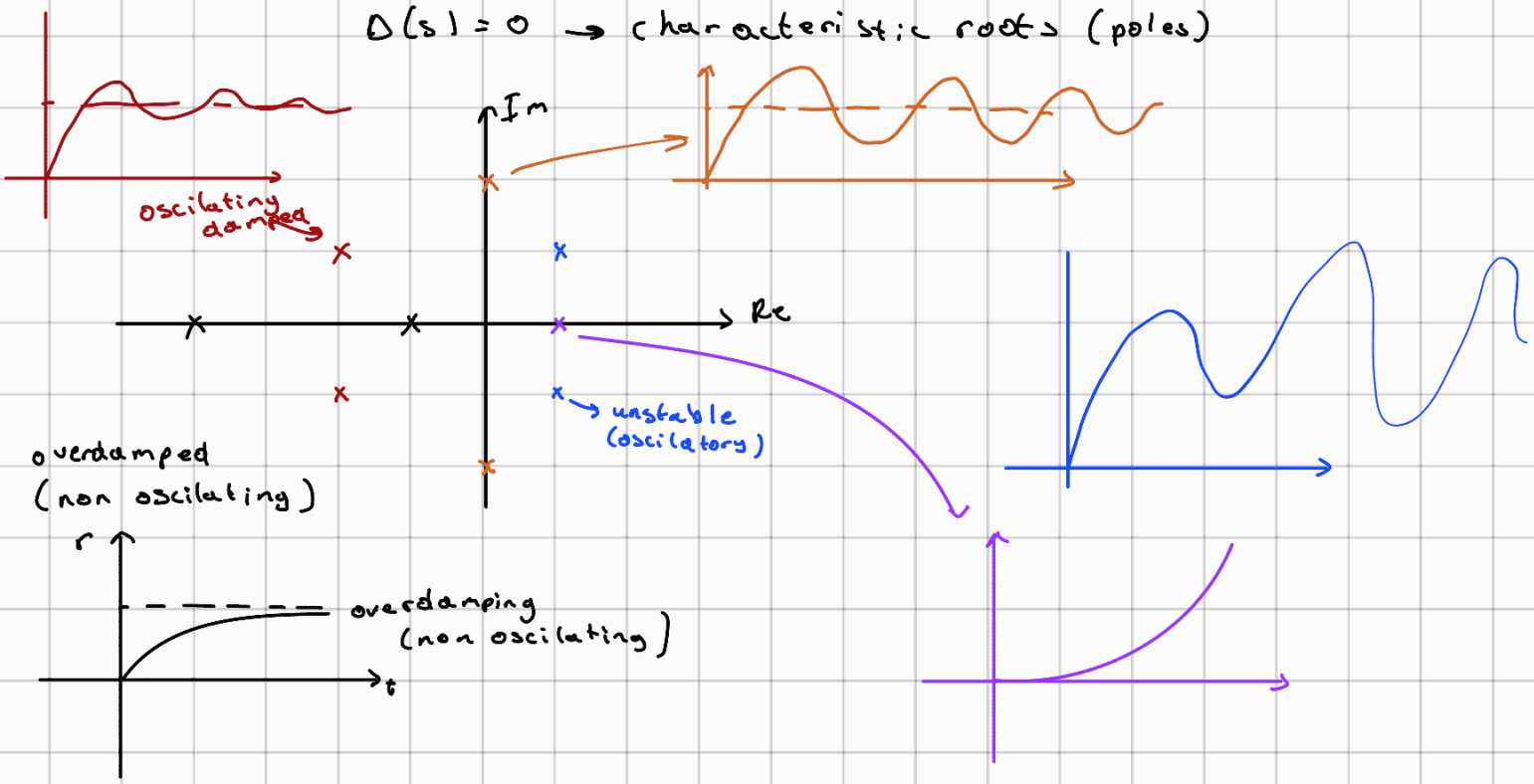
1 Quality of response

Closed loop TF = $\frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} = \frac{}{\Delta(s)}$

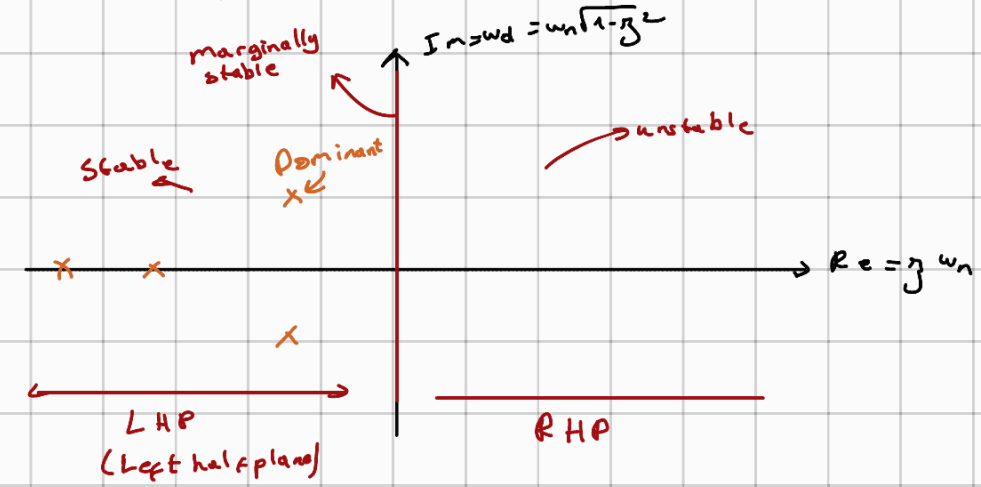
$\Delta(s) = 0 \rightarrow$ characteristic roots (poles)



2- Stability

This is a characktteristic of the system, and 'not' governed by the input signal

★ States of stability



Routh methode

Gives a way of determining stability state without the requirment of finding the poles

- No specific pole is given but the numbers of poles on the RHP is found.

$$\frac{C(s)}{R(s)} = \frac{b_0 s^m + b_1 s^{m-1} + \cdots + b_m}{a_0 s^n + a_1 s^{n-1} + \cdots + a_n} \qquad m \le n$$

(zeros: $b_0 s^m + \cdots + b_m$; poles: $a_0 s^n + \cdots + a_n$)

$\Delta(s)$ characterstic eq

* zero pole cancellation is not happening

$\Delta(s) = a_0 s^n + a_1 s^{n-1} + \cdots + a_{n-1} s + a_n$: polynomial of degree 'n'

if we can decompose it $(a_1 s^2 + a_2 s + a_3)$ and $(\beta_1 s + \beta_2)$ by multiplying them

$s + 2 = 0 \rightarrow s = -2$

$s - 2 = 0 \rightarrow s = +2$

so β_1 and β_2 should be positive

$\beta_1 = \frac{a_{n-1} a_n}{a_2}$

Requirement #1: all coefficients must be positive and nonzero!! (a_n)

Requirement #2: Construct the table of Routh

s^n	a_0	a_2	a_4	----	a_n
s^{n-1}	a_1	a_3	a_5	----	0
s^{n-2}	b_1	b_2	- ----	----	0 0
s^{n-3}	c_1	c_2	- - - - -	- - -	0
$\vdots$					
s^0	g_1	g_2	- ----		

$$b_1 = \frac{a_1 a_2 - a_0 a_3}{a_1}$$

$$b_2 = \frac{a_1 a_4 - a_0 a_5}{a_1}$$

$$\vdots$$

$$c_1 = \frac{b_1 a_3 - a_1 b_2}{b_1}$$

Conclusion: The number changes in sign = nubes of poles on the RHP (look to the first column)

Ex

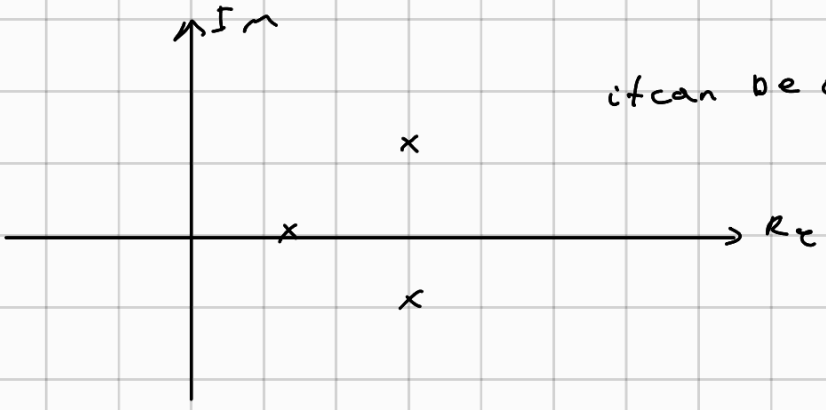
$s^4 + 2s^3 + 3s^2 + 4s + 5 = 0$

Pass the necessary condition

$$\frac{2 \times 3 - 1 \times 4}{2}$$

s^4	1	3	5
s^3	2	4	0
s^2	1	5	0
s^1	-6	0	
s^0	5		

2 changes of sign two poles on the RHP.



it can be oscilatory or not we dont know

Exception #1

The value of the cell in the first column at some specific row becomes zero

↳ just replace zero with $\varepsilon > 0$ and move on

$$\Delta(s) = s^3 - 3s + 2 = 0 \rightarrow s^3 + 0s^2 - 3s + 2 = 0$$

→ definitly unstable because there is negative coefficient

$\varepsilon > 0$

s^3	1	-3
s^2	ε	2
s^1	$\frac{-3\varepsilon - 2}{\varepsilon}$	0
s^0	2	

negative

we have got two roots on the jw axis.

Exception # 2

getting a full zero row!

$$s^5 + 2s^4 + 24s^3 + 48s^2 - 25s + 50 = 0$$

s^5	1	24	-25
s^4	2	48	-50
s^3	~~0~~ 8	~~0~~ 96	
s^2	24	-50	
s^1	112,7	0	
s^0	-50		

1 change in sign

→ 1 pole

it have to be real because complex pole have conjugate

construct an auxilary equation

$P(s) = 2s^4 + 48s^2 - 50$

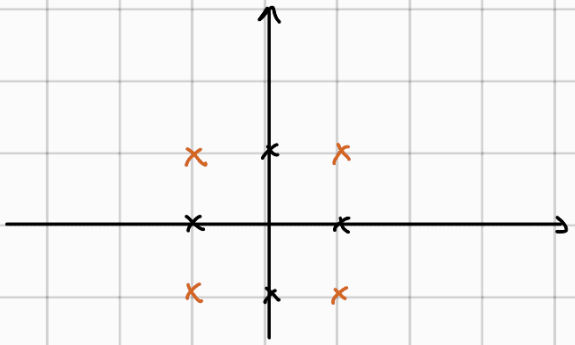
$\frac{dP}{ds} = 8s^3 + 96s$ → we have to use it

Solution of the auxilary eq. gives us some of the poles of the main syten

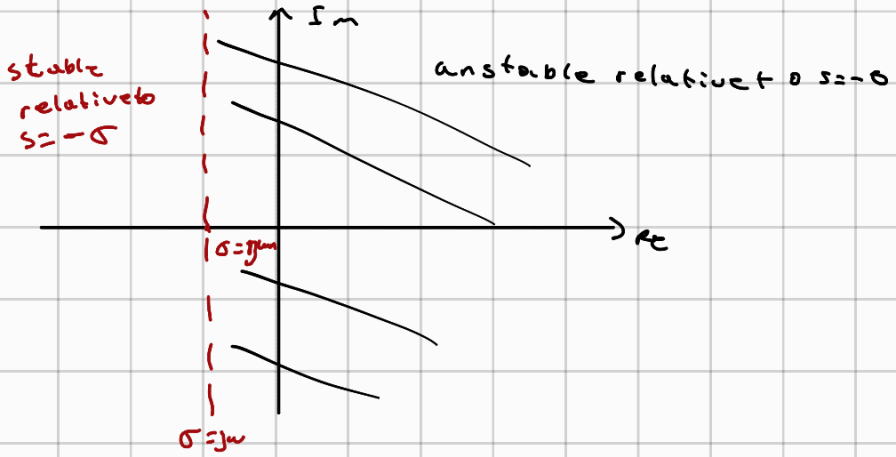
$$2s^4 + 48s^2 - 50 = 0 \rightarrow \begin{cases} s^2 = -25 \rightarrow s_{1,2} = \pm 5j \\ s^2 = 1 \rightarrow s_{3,4} = \pm 1 \end{cases}$$

case 1 of exception → at least two poles one on the imaginary axis

case 2 exception → some poles are symmetric with respect to the origin



Relative stability



$s + \sigma = \hat{s}$ at $s = -\sigma \Rightarrow \hat{s} = 0$

$\hookrightarrow s = \hat{s} - \sigma$

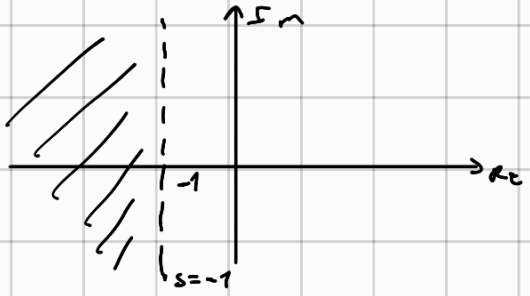
Ex $\dddot{c} + 6\ddot{c} + 11\dot{c} + kc = \dot{r} + r$

$$\frac{C(s)}{R(s)} = \frac{s^2 - 1}{s^3 + 6s^2 + 11s + k}$$

Ex: find possible k values so that all poles are in the shaded area?

$\Delta(s) = s^3 + 6s^2 + 11s + k = 0$

$s = \hat{s} - 1$



$$\Delta(\hat{s}) = (\hat{s}-1)^3 + 6(\hat{s}-1)^2 + 11(\hat{s}-1) + k = 0$$

$$= \hat{s}^3 + 3\hat{s}^2 + 2\hat{s} + k - 6 = 0$$

necessary condition

$k > 6$

$\hat{s}^3$	1	2	0
$\hat{s}^2$	3	$k-6$	0
$\hat{s}^1$	$\frac{6-(k-6)}{3}$	0	
$\hat{s}^0$	$k-6$		

all elements have to be positive

$$\frac{6-k+6}{3} = \frac{12-k}{3} > 0$$

$\Rightarrow$ ** $k < 12$

↳ $6 < k < 12$

Method #2 for the first exception case

$\begin{pmatrix} s_1 = +2 \\ d = \frac{1}{2} \end{pmatrix} \Rightarrow s = \frac{1}{d}$

This change of variable gives us the reciprocal of the root

$$s^5 + 2s^4 + 3s^3 + 6s^2 + 5s + 3 = 0$$

				if $\varepsilon > 0$	if $\varepsilon < 0$
s^5	1	3	5	+	+
s^4	2	6	3	+	+
s^3	~~0~~ $\varepsilon > 0$	7/2		+	−
s^2	$\frac{6\varepsilon - 7}{\varepsilon}$	3		−	+
s^1	$\frac{42\varepsilon - 49 - 6\varepsilon^2}{12\varepsilon - 14}$	0		+	+
s^0	3	0		+	+

Two poles are on the RHP

$$s = \frac{1}{\alpha}$$

$$\left(\frac{1}{\alpha}\right)^5 + 2\left(\frac{1}{\alpha}\right)^4 + 3\left(\frac{1}{\alpha}\right)^3 + 6\left(\frac{1}{\alpha}\right)^2 + 5\left(\frac{1}{\alpha}\right) + 3 = 0$$

$$\left(\frac{1}{\alpha}\right)^5 \left[1 + 2\left(\frac{1}{\alpha}\right)^{-1} + 3\left(\frac{1}{\alpha}\right)^{-2} + 6\left(\frac{1}{\alpha}\right)^{-3} + 5\left(\frac{1}{\alpha}\right)^{-4} + 3\left(\frac{1}{\alpha}\right)^{-5} \right] = 0$$

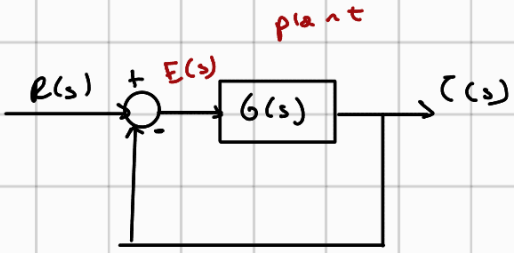
$$\Rightarrow 1 + 2\alpha + 3\alpha^2 + 6\alpha^3 + 5\alpha^4 + 3\alpha^5 = 0$$

α^5	3	6	2
α^4	5	3	1
α^3	4,2	1,4	
α^2	1,33	1	
α^1	-1,75	0	
α^0	1		

Solve this:

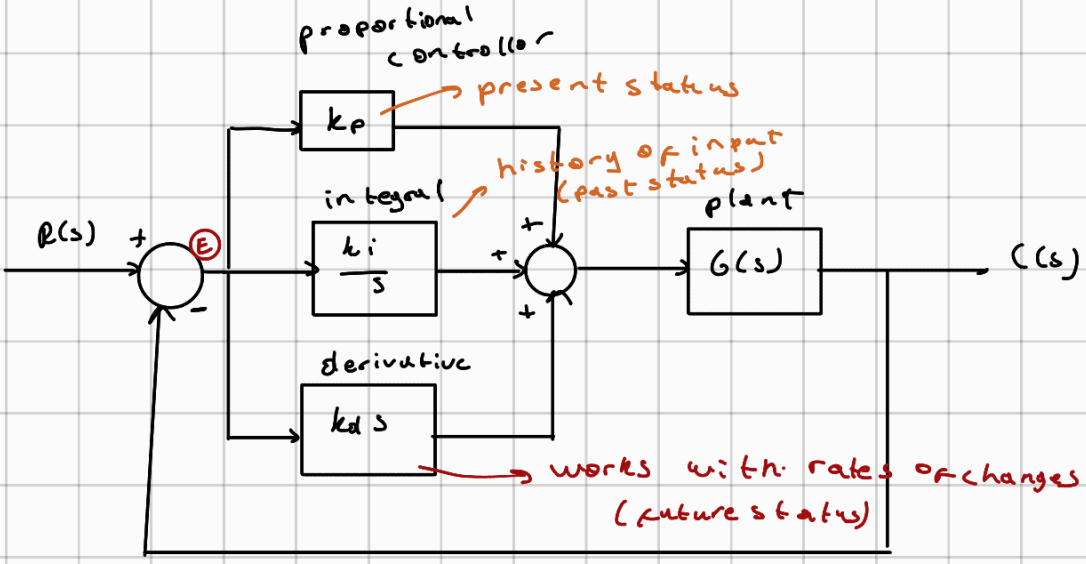
$$s^5 + 3s^4 + 5s^3 + 9s^2 + 9s^1 + 7s^0 + 3s + 2 = 0$$

Control actions

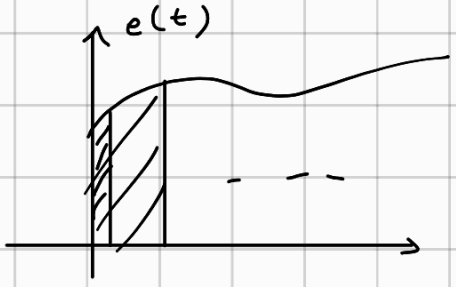


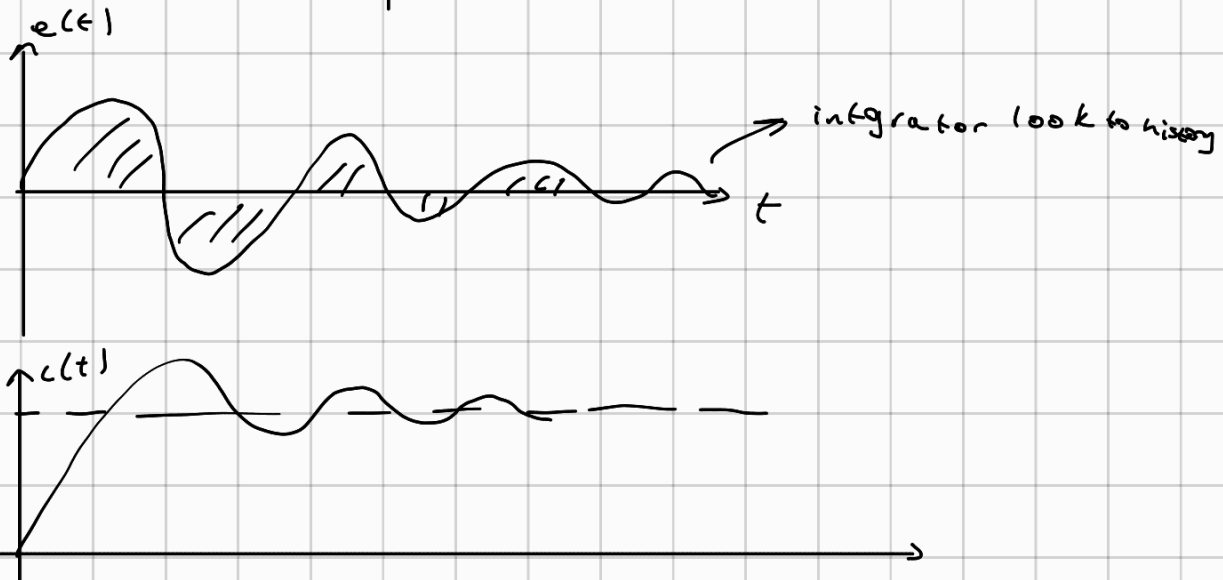
$E(s) = R(s) - C(s)$

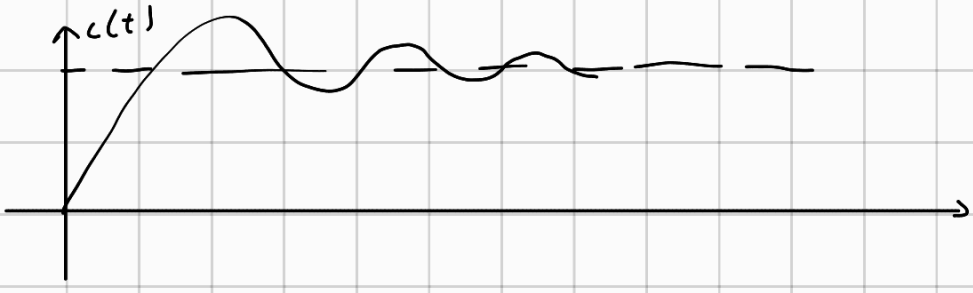
R(s) is the input and "can not" affect stability and accuracy of the system



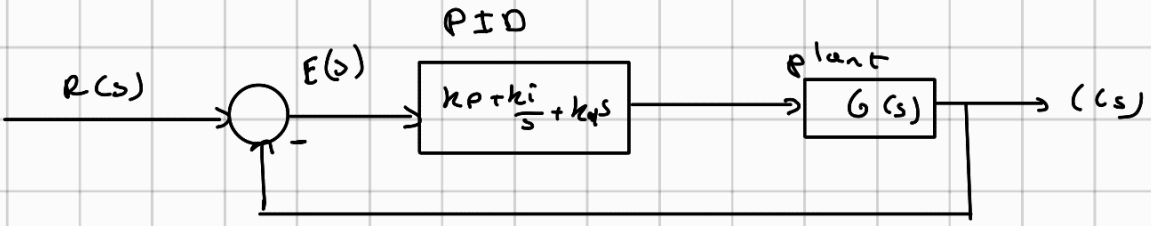
$k_i \int e(t)\,dt$







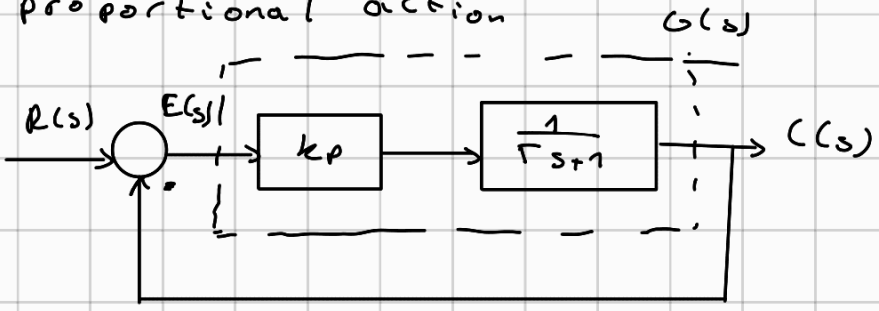
$k_d \dot{e}$ → rate of change of error



Derivative operation is always used with 'P' and 'I' operations

PD ✓ PI ✓ PID ✓
ID ✓
D ✗

1) proportional action → responsible for stability



$$G(s) = \frac{k_p}{\tau s+1}$$

$$E(s) = R(s) - G(s)$$

$$= R(s)\left(1 - \frac{C(s)}{R(s)}\right)$$

$$= R(s)\left(1 - \frac{G(s)}{1+G(s)}\right)$$

$$= R(s)\left(\frac{1}{1+G(s)}\right)$$

CLTF: $\frac{C(s)}{R(s)} = \frac{G(s)}{1+G(s)} = \frac{k_p}{\tau s+1+k_p}$

k_p is also responsible for the location of the pole

Steady state error

Final value theorem

$$e_{ss} = ? = e(\infty) = \lim_{t\to\infty} e(t) = \lim_{s\to 0} s\,E(s)$$

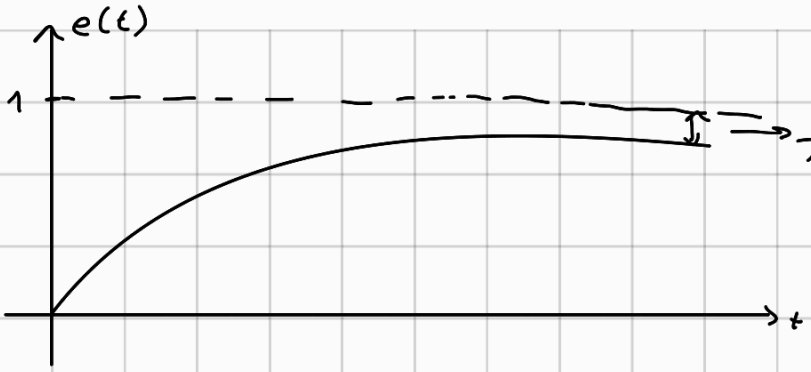
$$e_{ss} = \lim_{s\to 0} \frac{s\,R(s)}{1+G(s)}$$

If $R(s) = \frac{1}{s}$ step

$$e_{ss} = \frac{1}{1+G(0)}$$

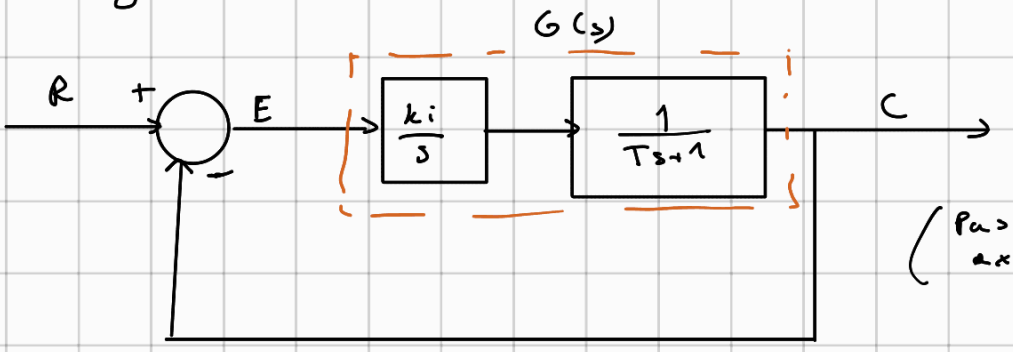
$$= \frac{1}{1+k_p}$$

$k_p \uparrow^{\infty} \Rightarrow e_{ss} \downarrow 0$

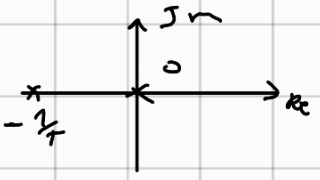


$\frac{1}{1+k_p}$ = offset

2) Integral action → responsible for error



(Pushing poles to jw axis make it more unstable)



$$G(s) = \frac{k_i}{s(Ts+1)}$$ (A pole added to the open loop function)

$$\frac{C(s)}{R(s)} = \frac{k_i}{s(Ts+1)+k_i}$$

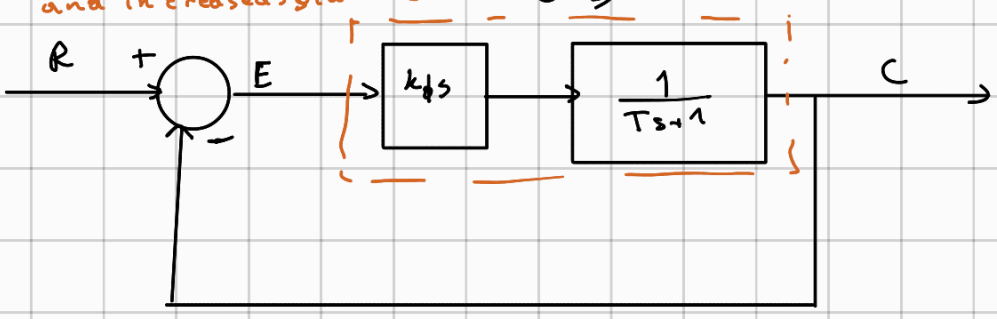
$$E(s) = R(s)\left(\frac{1}{1+G(s)}\right)$$

$$R(s) = \frac{1}{s}$$

$$e_{ss} = \lim_{s\to 0} s\,\frac{1}{s}\left(\frac{1}{1+G(s)}\right) = \frac{1}{1+G(0)} = 0$$

3) Derivative action (Responsible for sensitivity and increase stability in that sense)

$G(s) = \frac{k_d s}{Ts+1}$ OLTF

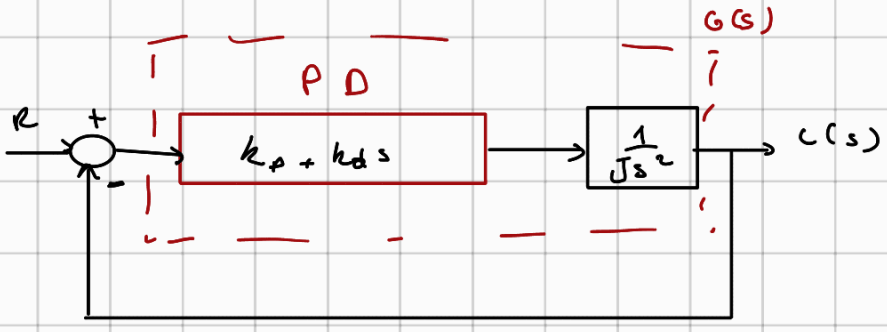


$$E(s) = e(s)\left(\frac{1}{1+G(s)}\right)$$



$$E(s) = R(s)\left(\frac{1}{1+G(s)}\right) = R(s)\left(\frac{Ts+1}{k_d s + Ts + 1}\right)$$

$$R(s) = \frac{1}{s} \Rightarrow e_{ss} = \lim_{s\to 0} s\left(\frac{1}{s}\right)\left(\frac{Ts+1}{k_d s + Ts + 1}\right) = 1$$



$$\frac{C(s)}{R(s)} = \frac{k_p + k_d s}{k_p + k_d s + J s^2} \qquad \Delta(s)$$

$$s^2 + \left(\frac{k_d}{J}\right)s + \left(\frac{k_p}{J}\right) = 0$$

$2\zeta\omega_n \qquad \omega_n^2$

$$\omega_n = \sqrt{\frac{k_p}{J}}$$

$$\zeta = \frac{1}{2\omega_n}\frac{k_d}{J} = \frac{1}{2}\sqrt{\frac{J}{k_p}}\frac{k_d}{J}$$

$$e^{-\zeta\omega_n t}\left(-\text{———}\right)$$

$$E(s) = R(s)\frac{1}{1+G(s)}$$

$$R(s) = \frac{1}{s}$$

$$e_{ss} = \lim_{s\to 0}\left(\frac{1}{s}\right)\frac{1}{1+G(0)}$$

$$G(s) = \frac{k_p + k_d s}{Js^2} = \infty$$

$$e_{ss} = 0$$